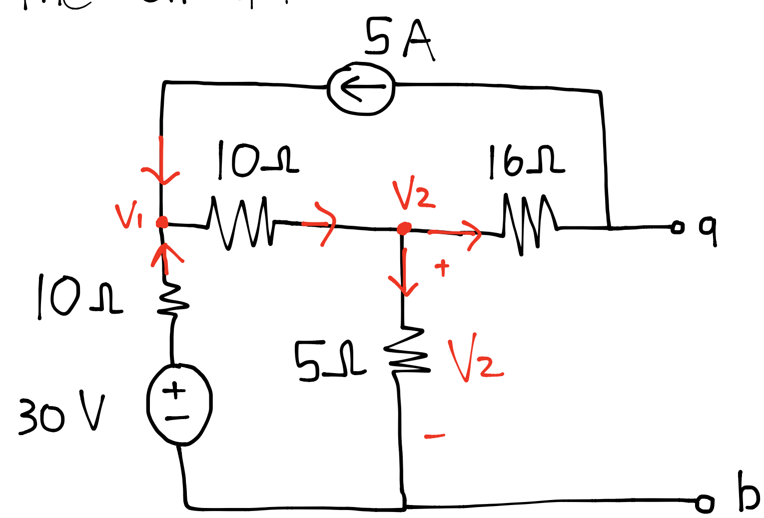
Tutorial 6

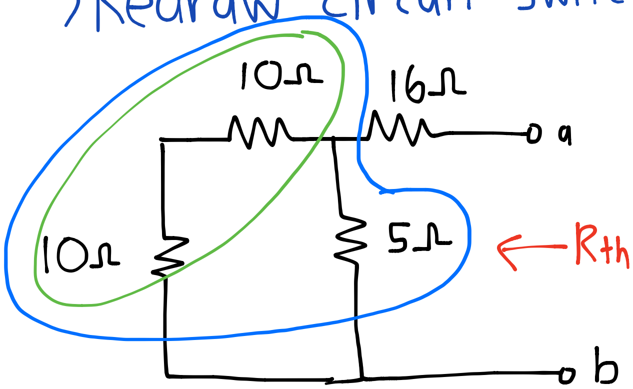
31 March 2016

4.39 Obtain the thevenin equival at terminals a-b of the circuit

* choose nodes before starting question



→ Redraw circuit switching off sources.



← R_{th}

→ Serie $= 10+10=20$

→ Parallel $= 20//5 = \frac{20\times5}{20+5} = 4$

→ Final series

$R_{th} = 16+4 = 20\Omega$

→ Look at sources for V_{TH}

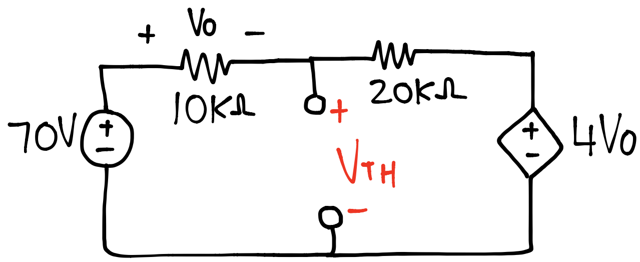
⇒ Node 1 : $\frac{30-V_1}{10} + 5 = \frac{V_1-V_2}{10} \longrightarrow 80 = 2V_1 - V_2 \ldots (1)$

⇒ Node 2 : $\frac{V_1-V_2}{10} = 5 + \frac{V_2}{5} \longrightarrow 100 = 2V_1 - 6V_2 \ldots (2)$

* Subtracting (1) from (2) $\Rightarrow 20 = -5V_2 \Rightarrow V_2 = \underline{-4\text{ V}}$

* KVL at V_{TH} : $-V_2 + 16\times5 + V_{TH} = 0 \Rightarrow \underline{V_{TH} = -84\text{ V}}$

4.40 Find the thevenin equivalent at terminals a-b



* Apply KVL over the whole loop

$$-70 + (10K + 20K) \times I + 4V_o = 0$$

$$\Rightarrow V_o = 10K \times I$$

$$\therefore 70 = 30K \times I + 4 \times 10K \times I$$

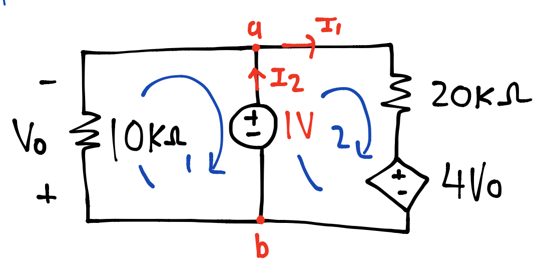
$$\Rightarrow 70 = 70K \times I \Rightarrow I = 1mA$$

* Apply KVL in inner loop to obtain V_{TH}

$$-70 + 10K \times I + V_{TH} = 0$$

$$V_{TH} = 70 - 10K \times 1m = \underline{60V}$$

* To find R_{TH} we must remove the source and apply a test source



* choose a 1V test Source to make the maths easier

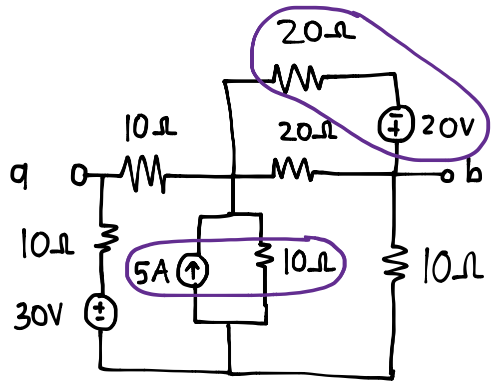
→ Notice that $V_o = -1V$

→ KVL in the 2nd loop $\quad -1 + 20K \times I_1 + 4V_o = 0 \quad \therefore I_1 = 0.25mA$

→ KCL at node a : $I_2 = I_1 + \frac{1V}{10k} = \underline{0.35mA}$

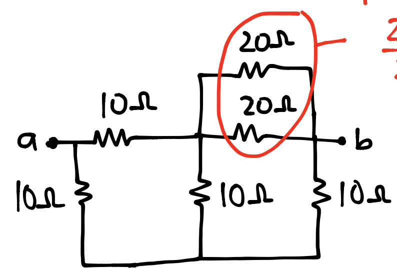
$$\therefore R_{TH} = \frac{1V \text{ (test source)}}{I_2} = \frac{1}{0.35 \times 10^{-3}} = \underline{2.857K\Omega}$$

4.42 Find the Thevenin equivalent between a and b



Switch off sources $\Longrightarrow$

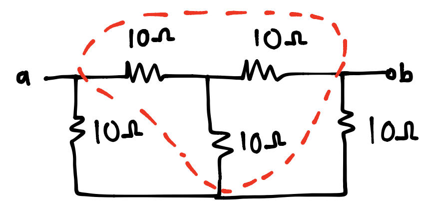
How to find R_{TH}



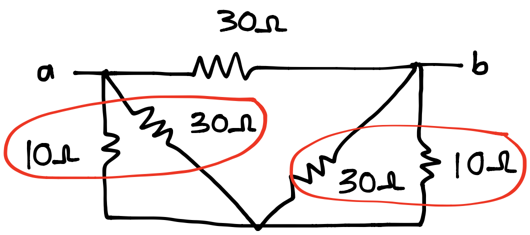
paralell $\frac{20\times20}{20+20} = 10$

$\Downarrow$

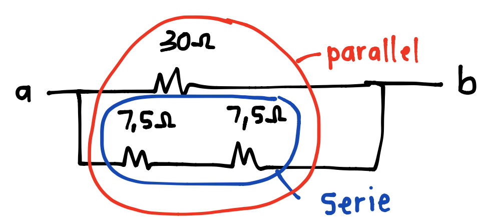
Simplify circuit



Y + Δ transform



$\Rightarrow$ paralell $\frac{10\times30}{10+30} = 7{,}5$

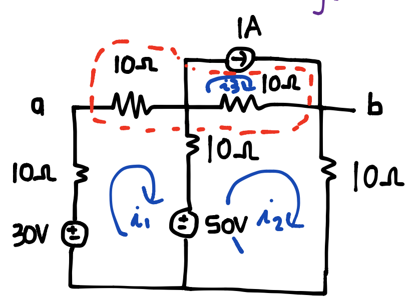


$\therefore R_{TH} = 30 // (7{,}5 + 7{,}5)$

$= \underline{10\Omega}$

*To find V_{TH} use source Transformation

① Current to Voltage transform



*NB

$i_3 = 1A$

* KVL in Loop 1:

$$-30 + 10i_1 + 10i_1 + 10(i_1 - i_2) + 50 = 0$$

$$30i_1 - 10i_2 = -20 \quad \dots (1)$$

* KVL in Loop 2:

$$-50 + 10(i_2 - i_1) + 10(i_2 - i_3) + 10i_2 = 0$$

$$-50 - 10i_1 + 30i_2 - 10 = 0$$

$$-5 - i_1 + 3i_2 - 1 = 0$$

$$i_1 = 3i_2 - 6 \quad \dots (2)$$

* Sub eq 2 into 1 $\therefore$ solve $\underline{i_2 = 2A}$

$\underline{i_1 = 0A}$

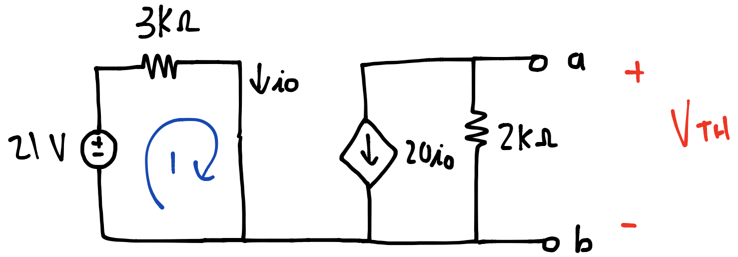
* NB: $V_{ab} = V_{TH}$ (Voltage across 2 10Ω resistors)

$$V_{th} = V_{10\Omega} + V_{10\Omega}$$

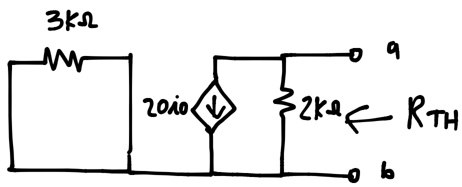
$$= 10i_1 + 10(i_2 - i_3)$$

$$= 10(0) + 10(2-1) = \underline{10V}$$

4.52 For the transistor model obtain the Thevenin equivalent at terminals a-b



* For the R_{TH} switch off sources



→ Due to no source in loop 1

$i_o = 0$ ∴ current source is also off

$R_{TH} = 2K\Omega$

* For V_{TH} look at the full circuit

→ KVL Loop 1 :

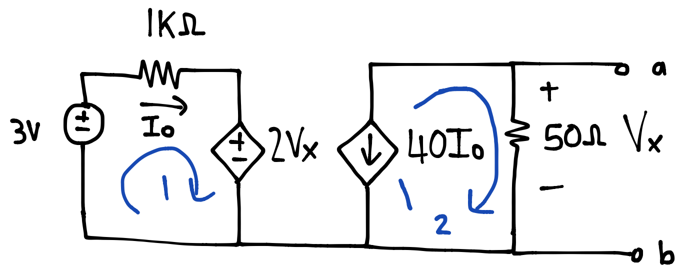
$21 = 3k \times i_o$ ∴ $i_o = 7mA$

$V_{TH} = -20 \times i_o \times 2K$

$= -20 \times 7m \times 2K$

$= -280V$

4.54 Find Thevenin equivalent between a-b



* $V_{TH} = V_x$ between a and b
* KVL Loop 1

$$-3 + 1K \times i_o + 2V_x = 0$$

* Loop 2 analysis

$$V_x = 50 \times (-40 I_o)$$

combining equations

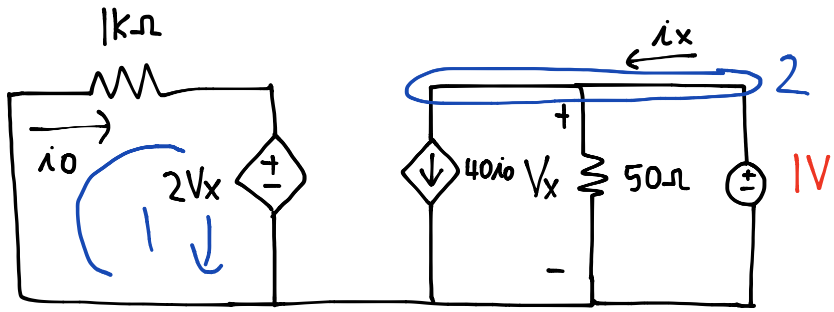
$$3 = 1000 i_o + 2[-50 \times 40 I_o]$$

$$\therefore I_o = -1mA$$

$$V_x = -2000(-1m) = 2$$

$$\therefore \underline{V_{TH} = 2V}$$

* To find R_{TH} swith off the source and implement test source (1V)



* Due to the test source in parallel with Vx

$\underline{V_x = 1V}$

⇒ Looking at loop 1

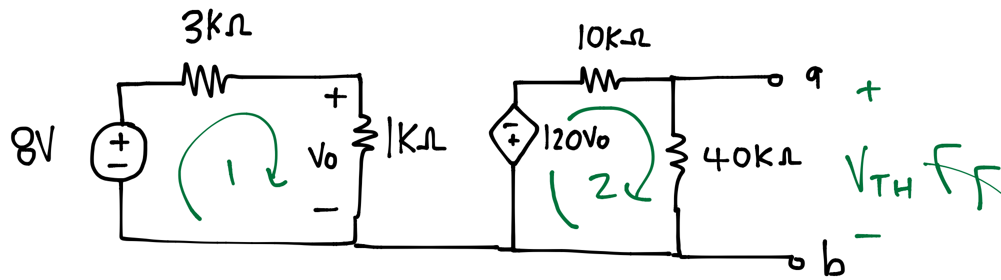
$$i_o = \frac{-2V_x}{1000} = -2mA$$

⇒ Looking at node 2 (KCL)

$$i_x = 40 i_o + \frac{V_x}{50} = -80m + \frac{1}{50}$$

$$= -60mA \therefore R_{TH} = \frac{1}{i_x} = \frac{1}{-0.06} = \underline{-16.67\Omega}$$

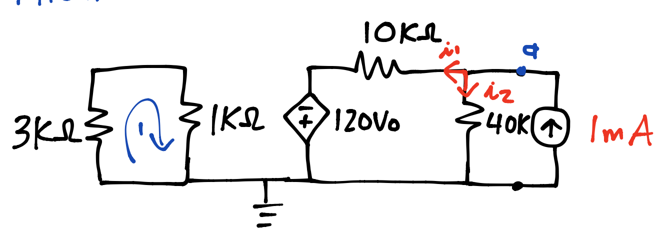
4.71 For the circuit, what resistor connected across a-b terminals will absorb maximum power from circuit? What is the power?



V_{TH}

* To find power we must find V_{TH} and R_{TH} at a and b

⇒ Redraw circuit: (with test source at a-b terminal)



* For V_{TH} look at the sources

* KCL at node a:

$$1 = \frac{V_a}{40} + \frac{V_a + 120V_o}{10} \Rightarrow 40 = 5V_a + 480V_o \quad \ldots (1)$$

* There is no Voltage source in Loop 1 ∴ $v_o = 0$ ... (2)

* Sub (2) in (1) ∴ $V_a = 8V$

* $R_{TH} = V_a / 1mA = 8K\Omega$ ∴ Resistance required

* Loop 1 ∴ $V_o = \frac{1}{4} \times 8 = 2V$ (Voltage division)

* Loop 2 ∴ $V_{TH} = \frac{40}{50} \times (-120V_o) = -192\,V$

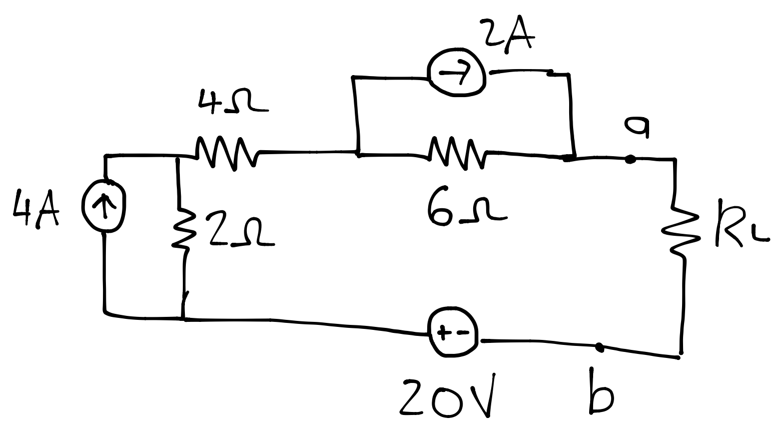
* power $= V_{TH}^2 / 4R_{TH} = (-192)^2 / (4 \times 8 \times 10^3) = 1.152$ watts

4.72 a) Obtain Thevenin equivalent at a-b

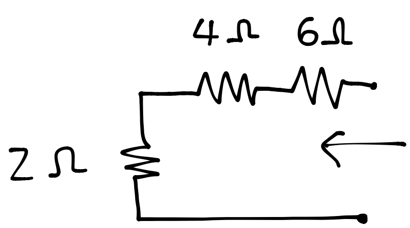
b) Calculate current in $R_L = 13\,\Omega$

c) Find R_L for maximum power deliverable to R_L

d) Determine the maximum power

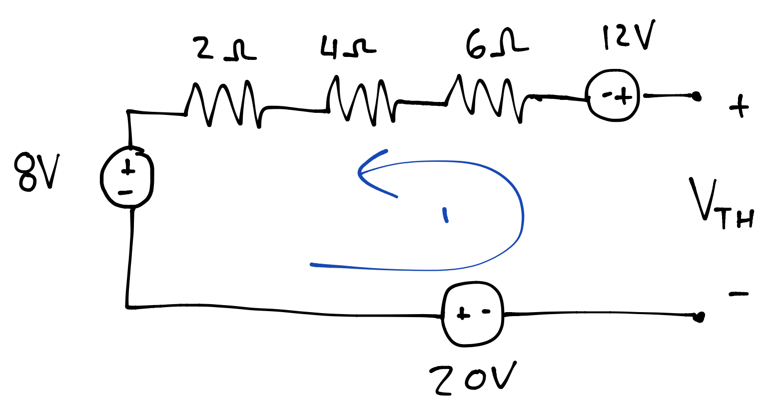


* a) Redraw circuit



$R_{TH} = 2 + 4 + 6 = \underline{12\,\Omega}$

Transform sources



* Add Voltages in loop

$$-V_{TH} + 12 + 8 + 20 = 0$$

$$\therefore \underline{V_{TH} = 40\,V}$$

*b) Current

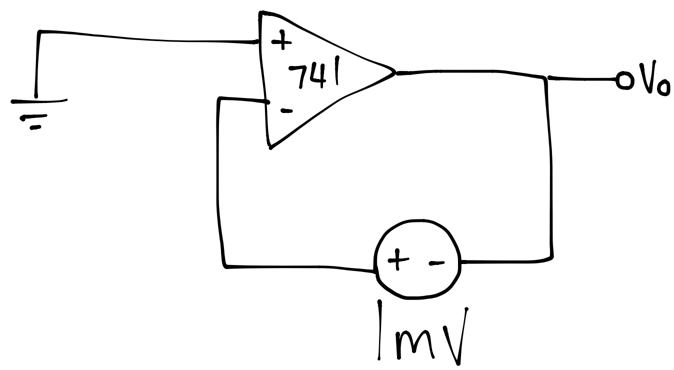
$$i = \frac{V_{TH}}{(R_{TH} + R)} = \frac{40}{(12+13)} = \underline{1.6A}$$

* c) For maximum power transfer

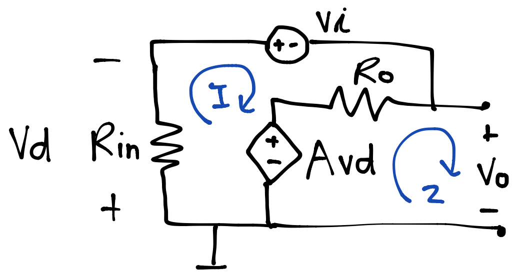
$$R_L = R_{TH} = \underline{12\,\Omega}$$

* d) $$\text{power} = \frac{V_{TH}^2}{4R_{TH}} = \frac{40^2}{(4 \times 12)} = \underline{33.33\text{ Watt}}$$

5.6 Using the same parameters as 741 opamp in Ex.1
Find V_o of the circuit below:



→ To obtain v_o we must look at the equivalent circuit:



* KVL in Loop 1:

$$(R_i + R_o)I + V_i + Av_d = 0 \quad \ldots (1)$$

* NB $V_d = R_i \times I$

⇒ Sub in (1). $V_i + (R_i + R_o + R_i A)I = 0$

→ Get I alone

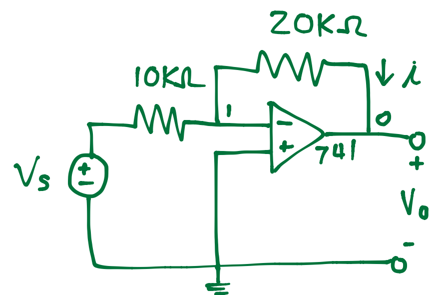
$$I = \frac{-V_i}{R_o + (1+A)R_i} \quad \ldots (2)$$

Ex.1

→ open-loop voltage gain $= 2 \times 10^5$

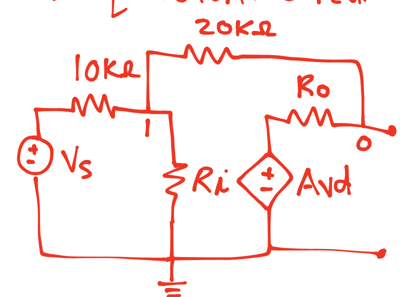
→ Input resistance $= 2M\Omega$

→ Output resistance $= 50\Omega$



? i when $v_s = 2V$

⇒ Equivalent circuit



→ KVL in Loop 2:

$$-Avd - R_oI + V_o = 0$$

$$V_o = Avd + R_oI$$

$$= (R_o + R_iA)I \quad \ldots \quad (3)$$

→ Sub (2) into (3)

$$V_o = (R_o + R_iA)\left[\frac{-V_i}{R_o + (1+A)R_i}\right]$$

$$= -V_i\left(\frac{R_o + R_iA}{R_o + (1+A)R_i}\right)V_i$$

⇒ Sub values $R_o = 50$ $R_i = 2M$ $A = 2\times10^5$ $V_i = 1\times10^{-3}$

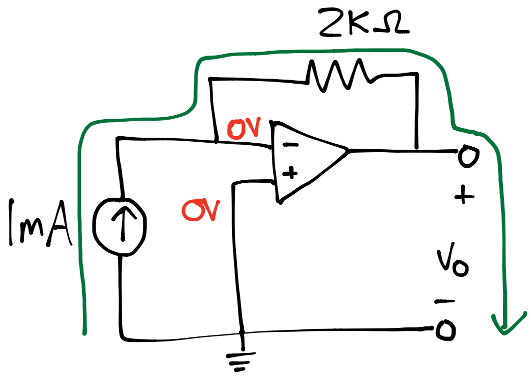
$$V_o = -\frac{(50 + 2\times10^6 \times 2\times10^5)\times10^{-3}}{50 + (1 + 2\times10^5)\times2\times10^6}$$

$$= \frac{-200{,}000 \times 2\times10^6}{200{,}001 \times 2\times10^6}$$

$$= -0.999995\,mV$$

5.8 Obtain V_o for each op amp

a)

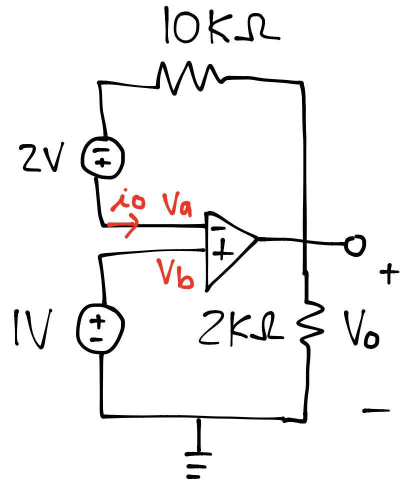


→ Due to inverting and noninverting terminals $V_a = V_b = 0V$

→ current flow

$$1mA = \frac{0 - V_o}{2K} \therefore \underline{V_o = -2V}$$

b)



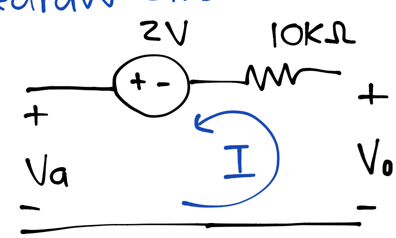
* $V_a = V_b = 1V$

→ No current at V_a

$i_o = 0$

* There is no current through 10KΩ

⇒ Redraw circuit



→ Looking at Loop

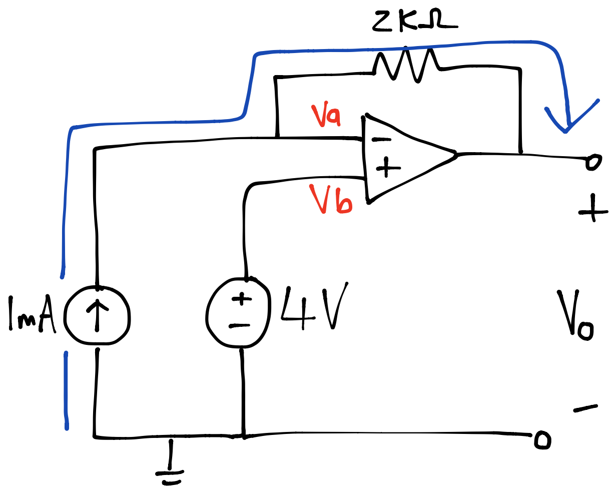
$$-V_a + 2 + V_o = 0$$

$$V_o = V_a - 2$$

$$= 1 - 2$$

$$\underline{= -1V}$$

5.9 Determine Vo for op amps

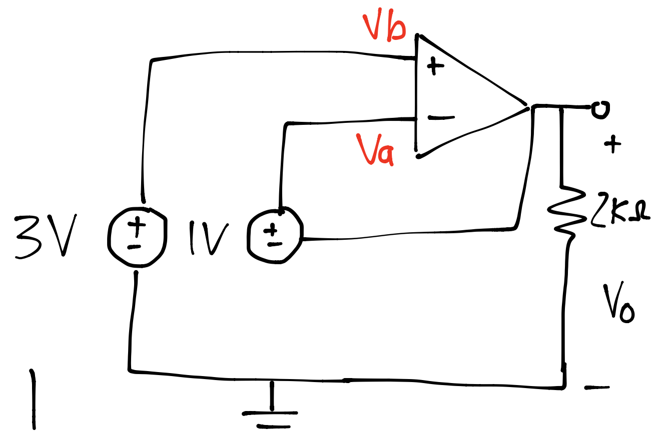


$* V_a = V_b = 4V$

⇒ Curret flow at inverting terminal:

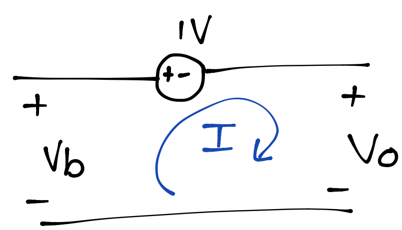
$$1mA = \frac{4 - V_o}{2K}$$

$$\therefore V_o = \underline{2V}$$



$V_a = V_b = 3V$

→ Redraw



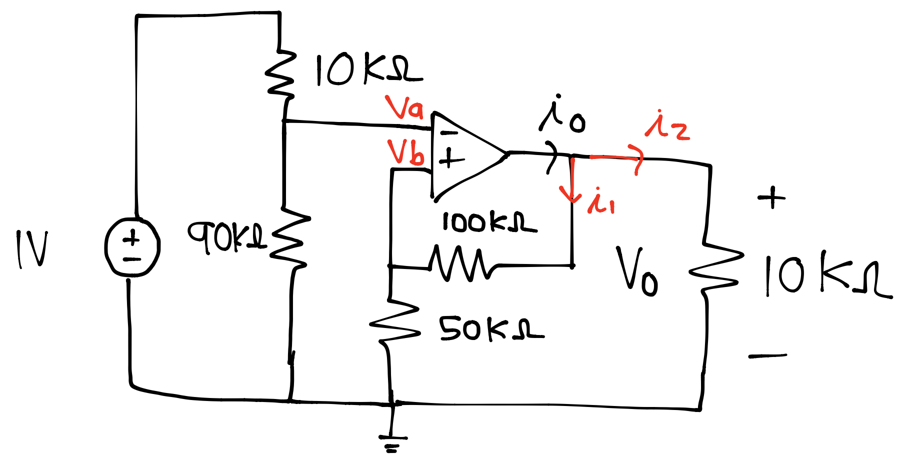
→ KVL in Loop

$$-V_b + 1 + V_o = 0$$

$$V_o = 3 - 1$$

$$= \underline{2V}$$

5.13 Find V_o and i_o in circuit



* Use voltage division

$$V_a = \frac{90}{100}(1) = \underline{0.9V}$$

$$V_b = \frac{50}{100} \times V_o = \underline{\frac{V_o}{3}}$$

* NB

$V_a = V_b$

$$\frac{V_o}{3} = 0.9$$

$$\therefore V_o = \underline{2.7V}$$

* Current at node:

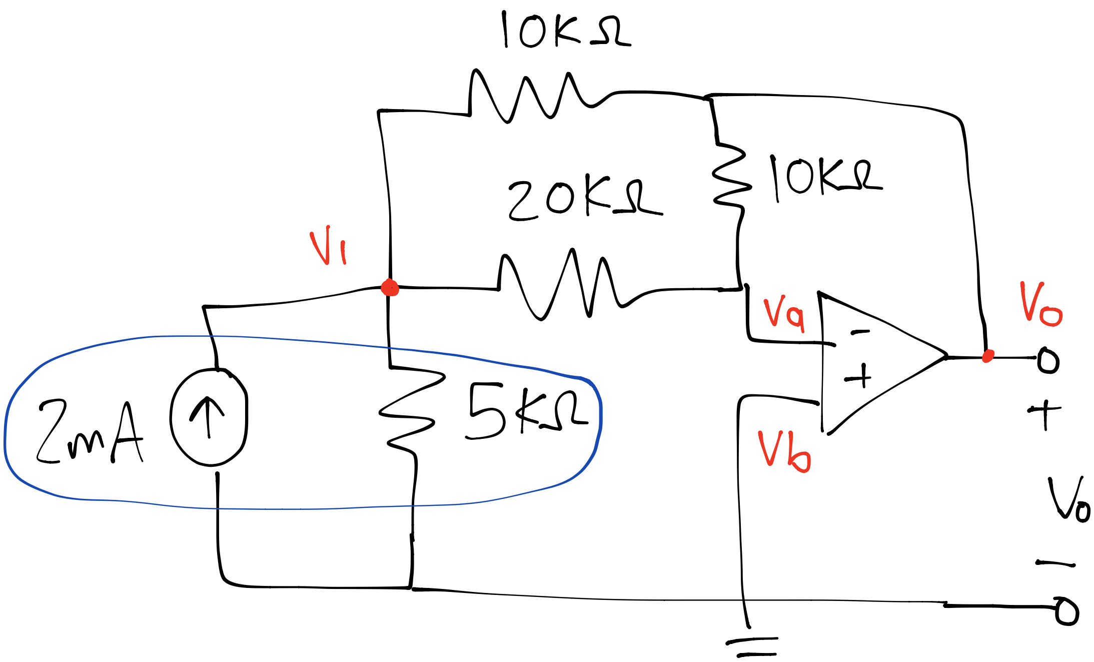
$$i_o = i_1 + i_2$$

$$= \frac{V_o}{150K} + \frac{V_o}{10K}$$

$$= 0.27m + 0.018m$$

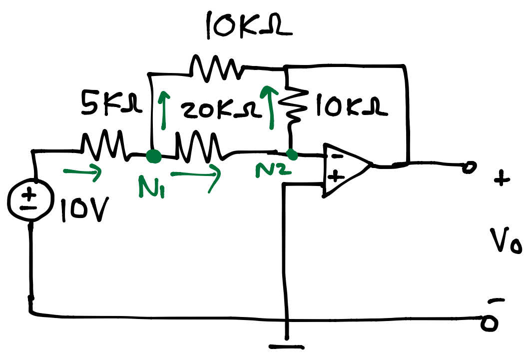
$$= 288uA$$

5.14 Determine the output voltage V_o



① Transform circuit for analysis

→ Redraw circuit:



• Nodal analysis

$$N_1: \frac{10 - V_1}{5} = \frac{V_1 - V_a}{20} + \frac{V_1 - V_o}{10}$$

NB $V_b = V_a = \underline{0V}$

$$\therefore 40 = 7V_1 - 2V_o \ldots (1)$$

$$N_2: \frac{V_1 - \overset{=0}{V_a}}{20} = \frac{\overset{=0}{V_a} - V_o}{10}$$

$$\therefore V_1 = -2V_o \ldots (2)$$

*Sub (1) and (2)

$$40 = 7(-2V_o) - 2V_o$$

$$\therefore V_o = \underline{-2.5V}$$